

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12407348
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Chuang; Fu-Chiang et al.

Touch detection circuit and operation device

Abstract

A touch detection circuit detects whether a key is being touched and includes a comparator, a compensation capacitor, and a voltage control circuit. The comparator includes an inverting input, a non-inverting input, and an output terminal. The compensation capacitor is coupled to the inverting input. The voltage control circuit provides an output voltage to the non-inverting input. In a calibration mode, the voltage control circuit adjusts the output voltage according to the voltage level of the output terminal.

Inventors: Chuang; Fu-Chiang (Hsinchu, TW), Tu; Chieh-Sheng (Hsinchu Science Park, TW)

Applicant: Nuvoton Technology Corporation (Hsinchu Science Park, TW)

Family ID: 90358755

Assignee: NUVOTON TECHNOLOGY CORPORATION (Hsinchu Science Park, TW)

Appl. No.: 18/321184

Filed: May 22, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240106433 A1	Mar. 28, 2024

Foreign Application Priority Data

TW	111136663	Sep. 28, 2022
----	-----------	---------------

Publication Classification

Int. Cl.: H03K17/00 (20060101); **G06F3/023** (20060101); **G06F3/044** (20060101); **H03K17/96** (20060101)

U.S. Cl.:

CPC **H03K17/962** (20130101); **G06F3/023** (20130101); **G06F3/044** (20130101);

Field of Classification Search

CPC: G06F (3/0354); G06F (3/044); G06F (3/0441); G06F (3/0442); G06F (3/0445); G06F (3/041); H03K (17/962)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10770024	12/2019	Ahn	N/A	G09G 3/3696
2011/0011717	12/2010	Lin	200/600	H03K 17/962
2013/0278509	12/2012	Chang	345/173	G06F 3/04166
2013/0342496	12/2012	Tsai	345/174	G06F 3/04182
2016/0070294	12/2015	Chung	331/8	G09G 5/12
2016/0188007	12/2015	Jung et al.	N/A	N/A
2018/0292923	12/2017	Peretz	N/A	G06F 3/041
2020/0210046	12/2019	Lee	N/A	G06F 3/0445
2021/0286462	12/2020	Wu	N/A	H03K 17/962
2023/0384885	12/2022	Kim	N/A	G06F 3/04164

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103513834	12/2013	CN	N/A
111258442	12/2019	CN	N/A
201102897	12/2010	TW	N/A
201539259	12/2014	TW	N/A

OTHER PUBLICATIONS

An Office Action in corresponding TW Application No. 111136663 dated May 29, 2023 is attached, 6 pages. cited by applicant

Primary Examiner: Lam; Tuan T

Attorney, Agent or Firm: MUNCY, GEISSLER, OLDS & LOWE, P.C.

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority of Taiwan Patent Application No. 111136663, filed on Sep. 28, 2022, the entirety of which is incorporated by reference herein.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The present invention relates to a detection circuit, and, in particular, to a detection circuit that determines whether a key is being touched.

Description of the Related Art

(3) With technological development, the types and functions of electronic devices have increased. Most electronic devices have an input interface that allows users to input information. Common input interfaces comprise keys and a mouse. When a key is touched, the electronic device performs a corresponding operation. However, the metal pad in the key can easily be disturbed by high-frequency signals, which can lead to the electronic devices misjudging when a key or button is being touched.

BRIEF SUMMARY OF THE INVENTION

(4) In accordance with an embodiment of the disclosure, a touch detection circuit detects whether a key is being touched and comprises a comparator, a compensation capacitor, and a voltage control circuit. The comparator comprises an inverting input, a non-inverting input, and an output terminal. The compensation capacitor is coupled to the inverting input. The voltage control circuit provides an output voltage to the non-inverting input. In a calibration mode, the voltage control circuit adjusts the output voltage according to the voltage level of the output terminal.

(5) In accordance with another embodiment of the disclosure, an operation device comprises a key and a touch detection circuit. The touch detection circuit is configured to determine whether the key is being touched and comprises a comparator, a compensation capacitor, and a voltage control circuit. The comparator comprises an inverting input, a non-inverting input, and an output terminal. The compensation capacitor is coupled to the inverting input. The voltage control circuit provides an output voltage to the non-inverting input. In a calibration mode, the voltage control circuit adjusts the output voltage according to the voltage level of the output terminal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention can be more fully understood by reading the subsequent detailed description and examples with references made to the accompanying drawings, wherein:

(2) FIG. 1 is a schematic diagram of an exemplary embodiment of an operation device according to various aspects of the present disclosure.

(3) FIG. 2 is a schematic diagram of an exemplary embodiment of a touch detection circuit according to various aspects of the present disclosure.

(4) FIG. 3 is a schematic diagram of an exemplary embodiment of a voltage control circuit according to various aspects of the present disclosure.

DETAILED DESCRIPTION OF THE INVENTION

(5) The present invention will be described with respect to particular embodiments and with reference to certain drawings, but the invention is not limited thereto and is only limited by the claims. The drawings described are only schematic and are non-limiting. In the drawings, the size of some of the elements may be exaggerated for illustrative purposes and not drawn to scale. The dimensions and the relative dimensions do not correspond to actual dimensions in the practice of the invention.

(6) FIG. 1 is a schematic diagram of an exemplary embodiment of an operation device according to various aspects of the present disclosure. The operation device **100** comprises a key **110** and a touch detection circuit **120**. The touch detection circuit **120** is configured to determine whether the key **110** is being touched. The kind of key **110** is not limited in the present disclosure. In one embodiment, the key **110** is a capacitive touch key. In FIG. 1, the capacitor C.sub.P indicates the parasitic capacitor of the key **110**. The capacitor C F indicates an additional touch capacitor

generated when the key **110** is touched. In some embodiments, the capacitor C.sub.P is generated by the electric field between the key **110** and other conductors, such as a ground layer, traces, product mechanisms, or any metals in the housing. The number of keys is not limited in the present disclosure. In other embodiments, the operation device **100** comprises more keys. In cases such as this, each key corresponds to a touch detection circuit or at least two keys share the same touch detection circuit.

(7) The touch detection circuit **120** comprises a comparator CMP, a compensation capacitor CCB and a voltage control circuit **121**. As shown in FIG. **1**, the comparator CMP comprises an inverting input CMPX_IN, a non-inverting input CMPX_IP, and an output terminal CMPX_OUT. The compensation capacitor CCB is coupled to the inverting input CMPX_IN. In this embodiment, the compensation capacitor CCB is a variable capacitor. The voltage control circuit **121** provides an output voltage V.sub.OUT to the non-inverting input CMPX_IP to control the voltage level of the output terminal CMPX_OUT.

(8) In a calibration mode, the voltage control circuit **121** adjusts the output voltage V.sub.OUT according to the voltage level of the output terminal CMPX_OUT. For example, when the voltage level of the output terminal CMPX_OUT is equal to the first specific level (e.g., a high level), the voltage control circuit **121** reduces the output voltage V.sub.OUT. After reducing the output voltage 'V_{Tom}', the voltage control circuit **121** detects the voltage level of the output terminal CMPX_OUT. When the voltage level of the output terminal CMPX_OUT is still equal to the first specific level, the voltage control circuit **121** re-reduces the output voltage 'V_{Tom}' until the voltage level of the output terminal CMPX_OUT is not equal to the first specific level. When the voltage level of the output terminal CMPX_OUT is not equal to the first specific level, the voltage control circuit **121** stops adjusting the output voltage 'V_{Tom}' and fixes the output voltage 'V_{Tom}'. At this time, the voltage of the non-inverting input CMPX_IP (i.e., the output voltage V.sub.OUT) is lower than the voltage of the inverting input CMPX_IN. Therefore, the voltage level of the output terminal CMPX_OUT is maintained to the second specific level (e.g., a lower level).

(9) Since the voltage level of the output terminal CMPX_OUT is maintained to the second specific level, the touch detection circuit **120** is capable of determining whether the key **110** is being touched according to the voltage level of the output terminal CMPX_OUT in an operation mode. For example, in an operation mode, when the key **110** is not touched, the voltage of the non-inverting input CMPX_IP is lower than the voltage of the inverting input CMPX_IN. Therefore, the voltage level of the output terminal CMPX_OUT is maintained to the second specific level. However, when the key **110** is being touched, the touch capacitor C_F is connected to the parasitic capacitor C.sub.P in parallel. At this time, the voltage of the inverting input CMPX_IN is reduced. When the voltage of the inverting input CMPX_IN is lower than the voltage of the non-inverting input CMPX_IP, the voltage level of the output terminal CMPX_OUT is not equal to the second specific level. Therefore, in the operation mode, the touch detection circuit **120** determines that the key **110** is being touched according to the voltage level of the output terminal CMPX_OUT.

(10) In some embodiments, the touch detection circuit **120** further comprises a processing circuit **122**. The processing circuit **122** sends an adjustment signal SA to adjust the capacitance of the compensation capacitor CCB. The present disclosure does not limit how the processing circuit **122** adjusts the capacitance of the compensation capacitor CCB. In one embodiment, the adjustment signal SA is a digital value. The range of the digital value is 0-255. When the adjustment signal SA is different digital values, the compensation capacitor CCB has different capacitance values. For example, when the value of the adjustment signal SA is 255, the capacitance of the compensation capacitor CCB has reached the upper limit, such as 4 pF. In other embodiments, before the processing circuit **122** adjusts the capacitance of the compensation capacitor CCB, the value of the adjustment signal SA is the initial value, such as **130**. In the operation mode, the processing circuit **122** adjusts the capacitance of the compensation capacitor CCB according to the voltage level of the output terminal CMPX_OUT. For example, when the voltage level of the output terminal

CMPX_OUT is not equal to the second specific level, it means that the key **110** is being touched. Therefore, the processing circuit **122** adjusts the capacitance of the compensation capacitor CCB to adjust the voltage of the inverting input CMPX_IN. In one embodiment, with increase of the capacitance of the compensation capacitor CCB, the voltage of the inverting input CMPX_IN is increased.

(11) When the voltage of the inverting input CMPX_IN is greater than the voltage of the non-inverting input CMPX_IP, the voltage level of the output terminal CMPX_OUT is equal to the second specific level, such as a lower level. The processing circuit **122** stops adjusting the compensation capacitor CCB. In one embodiment, the processing circuit **122** determines whether the variation of the capacitance of the compensation capacitor CCB has reached the threshold value. When the variation of the capacitance of the compensation capacitor CCB has not reached the threshold value, it means that the touch capacitor C F is caused by noise. However, when the variation of the capacitance of the compensation capacitor CCB reaches the threshold value, it means that the key **110** is indeed being touched. Therefore, the processing circuit **122** performs a corresponding operation. In one embodiment, the processing circuit **122** comprises a counter to count the duration of the output terminal CMPX_OUT being at a high level.

(12) In some embodiments, when the operation device **100** is powered on, the touch detection circuit **120** enters a calibration mode to ensure that the output terminal CMPX_OUT of the comparator CMP is maintained at a low level. Then, the touch detection circuit **120** enters an operation mode to determine whether the key **110** is being touched. In other embodiments, in the operation mode, the touch detection circuit **120** enters the calibration mode at regular time intervals. After the touch detection circuit **120** confirms that the output terminal CMPX_OUT of the comparator CMP is maintained at the low level, the touch detection circuit **120** re-enters the operation.

(13) In other embodiments, the compensation capacitor CCB receives the operation voltage V.sub.DD. In cases such as this, the operation voltage V.sub.DD is provided by a power source **123**. The compensation capacitor CCB is connected to the parasitic capacitor C.sub.P in parallel between the operation voltages V.sub.DD and V.sub.SS. When the key **110** is not touched, the capacitance of the compensation capacitor CCB and the parasitic capacitor C.sub.P divide the operation voltage V.sub.DD. At this time, the voltage of the inverting input CMPX_IN is greater than the voltage of the non-inverting input CMPX_IP. When the key **110** is touched, the touch capacitor C F is connected to the parasitic capacitor C.sub.P in parallel. Since the compensation capacitor CCB, the touch capacitor C F, and the parasitic capacitor C.sub.P divide the operation voltage V.sub.DD, the voltage of the inverting input CMPX_IN is reduced.

(14) FIG. 2 is a schematic diagram of an exemplary embodiment of the touch detection circuit according to various aspects of the present disclosure. When a scan signal SEN is at a low level, the voltages of the inverting input CMPX_IN and the non-inverting input CMPX_IP are equal to the operation voltage AVSS. When the scan signal SEN is changed from the low level to a high level, the touch detection circuit **120** determines whether the key **110** is being touched. When the key **110** is not being touched, the voltage of the inverting input CMPX_IN is greater than the voltage of the non-inverting input CMPX_IP. When the scan signal SEN is changed from the high level to the low level, the voltages of the inverting input CMPX_IN and the non-inverting input CMPX_IP are equal to the operation voltage AVSS. In one embodiment, the operation voltage V.sub.DD is fixed at a high voltage.

(15) FIG. 3 is a schematic diagram of an exemplary embodiment of the voltage control circuit **121** according to various aspects of the present disclosure. The voltage control circuit **121** comprises a digital-to-analog converter (DAC) **310** and a processing circuit **320**. The DAC **310** generates the output voltage V.sub.OUT according to the set value SV. In one embodiment, the DAC **310** comprises a control register **311**. The control register **311** is configured to store the set value SV. In some embodiment, the DAC **310** is coupled to the power source **123** to receive the operation

voltage V.sub.DD. In cases such as this, the operation voltage V.sub.DD is provided as a reference voltage V.sub.REF. The output voltage V.sub.OUT is in 0~V.sub.REF. In other words, the output voltage V.sub.OUT is less than or equal to the reference voltage V.sub.REF.

(16) The processing circuit **320** adjusts the set value SV stored in the control register **311** according to the voltage level of the output terminal CMPX_OUT. In one embodiment, the processing circuit **320** comprises a register (not shown) to store firmware. The processing circuit **320** executes the firmware to write a corresponding set value SV into the control register **311**. In the calibration mode, the processing circuit **320** detects the voltage level of the output terminal CMPX_OUT. When the voltage level of the output terminal CMPX_OUT is equal to the first specific level (e.g., a high level), the processing circuit **320** gradually reduces or increases the set value SV. When this happens, the DAC **310** generates different output voltages V.sub.OUT according to different set values SV.

(17) In some embodiments, the key **110** is easily interfered by high-frequency signals. Therefore, before the touch detection circuit **120** determines whether the key **110** is being touched, the touch detection circuit **120** first performs a calibration operation to set the voltage level of the output terminal CMPX_OUT of the comparator CMP being at a low level. In this embodiment, since the output voltage V.sub.OUT covers a wide range, it can ensure that the voltage level of the output terminal CMPX_OUT of the comparator CMP is a low level when the key is not touched. Additionally, the voltage control circuit **121** is used to calibrate the voltage level of the output terminal CMPX_OUT, the non-inverting input CMPX_IP does not need to receive an external calibration signal.

(18) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein. It will be understood that although the terms “first,” “second,” etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another.

(19) While the invention has been described by way of example and in terms of the preferred embodiments, it should be understood that the invention is not limited to the disclosed embodiments. On the contrary, it is intended to cover various modifications and similar arrangements (as would be apparent to those skilled in the art). Therefore, the scope of the appended claims should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements.

Claims

1. A touch detection circuit for detecting whether a key is being touched, comprising: a comparator comprising an inverting input, a non-inverting input, and an output terminal; a compensation capacitor coupled to the inverting input; a first processing circuit adjusting the capacitance of the compensation capacitor according to a voltage level of the output terminal; and a voltage control circuit providing an output voltage to the non-inverting input, wherein in a calibration mode, the voltage control circuit adjusts the output voltage according to the voltage level of the output terminal.
2. The touch detection circuit as claimed in claim 1, wherein in the calibration mode and in response to the voltage level of the output terminal being equal to a first specific level, the voltage control circuit reduces the output voltage until the voltage level of the output terminal is not equal to the first specific level.
3. The touch detection circuit as claimed in claim 1, wherein the voltage control circuit comprises:

- a digital-to-analog converter generating the output voltage according to a set value; and a second processing circuit adjusting the set value according to the voltage level of the output terminal.
4. The touch detection circuit as claimed in claim 3, wherein the digital-to-analog converter comprises: a control register configured to store the set value.
 5. The touch detection circuit as claimed in claim 3, further comprising: a power source configured to provide an operation voltage to the compensation capacitor and the digital-to-analog converter.
 6. The touch detection circuit as claimed in claim 5, wherein the maximum level of the output voltage is equal to the operation voltage.
 7. The touch detection circuit as claimed in claim 1, wherein in an operation mode, the voltage control circuit stops adjusting the output voltage.
 8. The touch detection circuit as claimed in claim 7, wherein in the operation mode and in response to the voltage level of the output terminal not being equal to a second specific level, the first processing circuit increases the capacitance of the compensation capacitor until the voltage level of the output terminal is equal to the second specific level.
 9. The touch detection circuit as claimed in claim 8, wherein: the first processing circuit determines whether a variation of the capacitance of the compensation capacitor has reached a threshold value, and in response to the variation of the capacitance of the compensation capacitor reaching the threshold value, it is determined that the key is being touched.
 10. The touch detection circuit as claimed in claim 1, wherein: in response to the key not being touched, a voltage of the inverting input is higher than a voltage of the non-inverting input, and in response to the key being touched, the voltage of the inverting input is lower than the voltage of the non-inverting input.
 11. An operation device comprising: a key; and a touch detection circuit configured to determine whether the key is being touched and comprising: a comparator comprising an inverting input, a non-inverting input, and an output terminal; a compensation capacitor coupled to the inverting input; a first processing circuit adjusting the capacitance of the compensation capacitor according to a voltage level of the output terminal; and a voltage control circuit providing an output voltage to the non-inverting input, wherein in a calibration mode, the voltage control circuit adjusts the output voltage according to the voltage level of the output terminal.
 12. The operation device as claimed in claim 11, wherein in the calibration mode and in response to the voltage level of the output terminal being equal to a first specific level, the voltage control circuit reduces the output voltage until the voltage level of the output terminal is not equal to the first specific level.
 13. The operation device as claimed in claim 11, wherein the voltage control circuit comprises: a digital-to-analog converter generating the output voltage according to a set value; and a second processing circuit adjusting the set value according to the voltage level of the output terminal.
 14. The operation device as claimed in claim 13, wherein the digital-to-analog converter comprises: a control register configured to store the set value.
 15. The operation device as claimed in claim 13, further comprising: a power source configured to provide an operation voltage to the compensation capacitor and the digital-to-analog converter.
 16. The operation device as claimed in claim 15, wherein the maximum level of the output voltage is equal to the operation voltage.
 17. The operation device as claimed in claim 11, wherein in an operation mode, the voltage control circuit stops adjusting the output voltage.
 18. The operation device as claimed in claim 17, wherein in the operation mode and in response to the voltage level of the output terminal not being equal to a second specific level, the first processing circuit increases the capacitance of the compensation capacitor until the voltage level of the output terminal is equal to the second specific level.
 19. The operation device as claimed in claim 18, wherein: the first processing circuit determines whether a variation of the capacitance of the compensation capacitor has reached the threshold

value, and in response to the variation of the capacitance of the compensation capacitor reaching the threshold value, it is determined that the key is being touched.

20. The operation device as claimed in claim 11, wherein: in response to the key not being touched, a voltage of the inverting input is higher than a voltage of the non-inverting input, and in response to the key being touched, the voltage of the inverting input is lower than the voltage of the non-inverting input.
